

IMAT REPAIR DRILL — Repeat Offenders

70 questions · 100 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · Repair drill

Section A - Respiration & Bioenergetics (Q1-12)

1. In a eukaryotic cell, the enzymes that catalyse the citric acid (Krebs) cycle are located mainly in the:
A) intermembrane space
B) cytosol
C) mitochondrial matrix
D) inner mitochondrial membrane
E) outer mitochondrial membrane
2. Which of the following processes occurs in the mitochondrial matrix?
A) reduction of O_2 to water at complex IV
B) electron transfer between respiratory complexes I and IV
C) conversion of glucose to pyruvate
D) ATP synthesis driven by a proton gradient
E) oxidation of acetyl-CoA to CO_2 in the citric acid cycle
3. In human cells, ATP synthase is located in the:
A) outer mitochondrial membrane
B) intermembrane space
C) cytosol
D) inner mitochondrial membrane
E) mitochondrial matrix
4. Cyanide blocks complex IV, the last complex of the electron transport chain. What is the most immediate effect on ATP synthase?
A) It stops making ATP because the H^+ gradient across the inner membrane dissipates.
B) It continues making ATP because it phosphorylates ADP by substrate-level phosphorylation.
C) It makes more ATP because protons accumulate in the matrix.
D) It keeps working normally because ATP synthase is not part of the electron transport chain.
E) It switches to using electrons taken directly from glucose.
5. During the complete aerobic oxidation of one glucose molecule, carbon dioxide is released:
A) during glycolysis only
B) during pyruvate oxidation and the citric acid cycle
C) during glycolysis and the electron transport chain
D) during substrate-level phosphorylation only
E) during the electron transport chain only
6. A working muscle cell temporarily carries out only glycolysis followed by lactic acid fermentation. How many molecules of CO_2 are released per molecule of glucose consumed?
A) 0 **B)** 1 **C)** 2 **D)** 3 **E)** 6
7. Most of the carbon dioxide carried in human blood is transported:
A) as carbonic acid bound to albumin
B) dissolved as CO_2 in the plasma
C) as carbaminohemoglobin bound to hemoglobin
D) attached to the heme iron of hemoglobin
E) as bicarbonate ions (HCO_3^-) in the plasma
8. Approximately what percentage of the carbon dioxide in human blood is transported as bicarbonate ions (HCO_3^-)?
A) about 23% **B)** about 7% **C)** about 97% **D)** about 70% **E)** about 50%

9. When a cell's ATP concentration is high, the rate of glycolysis falls. This happens because ATP:
- A) competes with fructose-6-phosphate for the active site of phosphofructokinase-1
 - B) activates phosphofructokinase-1 and accelerates glucose breakdown
 - C) binds allosterically to phosphofructokinase-1 and lowers its activity
 - D) competitively inhibits pyruvate kinase
 - E) irreversibly blocks the active site of hexokinase
10. In a eukaryotic cell, complete oxidation of one glucose molecule yields roughly 30-32 ATP. Most of this ATP is produced by:
- A) lactic acid fermentation inside the mitochondria
 - B) oxidative phosphorylation powered by the electron transport chain
 - C) substrate-level phosphorylation in the citric acid cycle
 - D) direct phosphorylation during pyruvate oxidation
 - E) substrate-level phosphorylation in glycolysis
11. During skeletal muscle contraction, the myosin heads attach directly to:
- A) calcium ions
 - B) the Z disc
 - C) actin
 - D) troponin
 - E) tropomyosin
12. When oxygen is unavailable, yeast and muscle cells run fermentation chiefly in order to:
- A) produce NADH for the electron transport chain
 - B) convert NAD⁺ into NADP⁺
 - C) remove excess NAD⁺ from the cytosol
 - D) generate extra ATP beyond that made in glycolysis
 - E) regenerate NAD⁺ so that glycolysis can continue

Section B - Body Systems (Q13-20)

13. Bile, which is produced by the liver and stored in the gallbladder, assists fat digestion primarily by...
- A) emulsifying large fat globules into tiny droplets, increasing the surface area available to lipase
 - B) hydrolysing triglycerides into fatty acids and glycerol
 - C) lowering the pH of chyme so that pancreatic enzymes can act
 - D) secreting lipase, the enzyme that chemically digests fats
 - E) activating pepsinogen into pepsin in the stomach
14. During a normal inhalation, which sequence of events occurs?
- A) The diaphragm relaxes and domes upward, thoracic volume decreases, and pressure inside the lungs rises above atmospheric.
 - B) The diaphragm contracts and flattens, thoracic volume increases, and pressure inside the lungs rises above atmospheric.
 - C) The diaphragm relaxes and domes upward, thoracic volume increases, and pressure in the lungs falls below atmospheric.
 - D) The diaphragm contracts and flattens, thoracic volume increases, and pressure in the lungs falls below atmospheric, drawing air in.
 - E) The diaphragm contracts and flattens, thoracic volume decreases, and pressure inside the lungs rises.
15. During quiet breathing at rest, expiration is best described as...
- A) a passive process in which the diaphragm contracts while thoracic volume stays constant
 - B) a passive process in which the diaphragm relaxes and domes upward, reducing thoracic volume
 - C) an active process in which the abdominal muscles contract forcibly to empty the lungs
 - D) an active process in which the diaphragm contracts and flattens to push air out
 - E) a passive process in which the diaphragm relaxes while thoracic volume increases

- 16.** Oxygenated blood is delivered to the systemic circulation by which route?
- A)** From the left ventricle into the pulmonary artery
 - B)** From the left ventricle into the aorta
 - C)** From the right atrium into the vena cava
 - D)** From the left atrium into the pulmonary vein
 - E)** From the right ventricle into the aorta
- 17.** The semilunar valves of the heart prevent...
- A)** blood from flowing back from the atria into the venae cavae and pulmonary veins
 - B)** blood from flowing back from the ventricles into the atria
 - C)** blood from flowing from the atria into the ventricles during diastole
 - D)** blood from flowing back from the aorta and pulmonary artery into the ventricles
 - E)** blood from flowing from the pulmonary veins into the left atrium
- 18.** The urinary bladder must tolerate repeated stretching as it fills with urine. Which epithelium lines it?
- A)** Stratified squamous epithelium
 - B)** Simple squamous epithelium
 - C)** Simple columnar epithelium
 - D)** Ciliated pseudostratified epithelium
 - E)** Transitional epithelium
- 19.** The kidney tubules and many secretory glands are lined mainly by which epithelium?
- A)** Simple columnar epithelium
 - B)** Simple squamous epithelium
 - C)** Simple cuboidal epithelium
 - D)** Stratified squamous epithelium
 - E)** Transitional epithelium
- 20.** Ciliated epithelium lining the trachea serves mainly to...
- A)** sweep mucus with trapped dust and microbes upward toward the pharynx
 - B)** contract rhythmically to narrow the airway during coughing
 - C)** secrete the mucus that traps inhaled particles
 - D)** absorb oxygen directly from the air passing through the trachea
 - E)** provide the main surface for gas exchange with the blood

Section C - Cell Division & Cell Biology (Q21-30)

- 21.** Nondisjunction in meiosis I, meaning the failure of homologous chromosomes to separate from each other, occurs during:
- A)** telophase I **B)** prophase I **C)** anaphase I **D)** metaphase I **E)** anaphase II
- 22.** In a human primary spermatocyte ($2n = 46$) undergoing meiosis II, nondisjunction would be the failure of which structures to separate?
- A)** chromosomes from the spindle poles
 - B)** bivalents at the metaphase plate
 - C)** homologous chromosomes of each pair
 - D)** sister chromatids of each chromosome
 - E)** non-sister chromatids at the chiasmata
- 23.** Independent assortment of maternal and paternal chromosomes results from their random orientation at the metaphase plate. This alignment of homologous pairs occurs during:
- A)** prophase I of meiosis
 - B)** metaphase I of meiosis
 - C)** metaphase of mitosis
 - D)** metaphase II of meiosis
 - E)** telophase I of meiosis

- 24.** Crossing over, the exchange of corresponding segments between homologous chromosomes, takes place during:
- A)** prophase I **B)** anaphase I **C)** metaphase I **D)** prophase II **E)** telophase II
- 25.** Glucose and galactose differ in the spatial arrangement of the hydroxyl group at a single chiral carbon, C-4. They are therefore classified as:
- A)** cis-trans isomers
B) structural isomers
C) enantiomers
D) anomers
E) epimers
- 26.** Tay-Sachs disease is caused by a defective enzyme, hexosaminidase A, so an undegraded membrane lipid (GM2 ganglioside) accumulates. In which organelle does the lipid build up?
- A)** smooth endoplasmic reticulum
B) peroxisome
C) lysosome
D) Golgi apparatus
E) mitochondrion
- 27.** The accumulation of misfolded proteins inside the endoplasmic reticulum triggers the unfolded protein response. This misfolded-protein build-up is sensed in the:
- A)** smooth endoplasmic reticulum
B) cytosol
C) Golgi apparatus
D) rough endoplasmic reticulum
E) lysosome
- 28.** Parathyroid hormone raises blood calcium concentration. Parathyroid hormone does all of the following EXCEPT:
- A)** stimulate the release of calcitonin from the thyroid gland
B) increase reabsorption of calcium in the kidney
C) stimulate osteoclast-mediated bone resorption
D) increase phosphate excretion in the urine
E) promote activation of vitamin D
- 29.** An intravenous drip delivers 0.9% saline, whose osmolarity matches that of blood plasma. Red blood cells suspended in this solution will:
- A)** shrink because the solution is hypotonic
B) swell because the solution is hypertonic
C) shrink and crenate because the solution is hypertonic
D) swell and possibly lyse because the solution is hypotonic
E) keep their normal shape because the solution is isotonic
- 30.** A catastrophe reduces a large population to a few randomly surviving individuals, which then rebuild the population. The restored population shows greatly reduced genetic diversity. This is an example of:
- A)** sexual selection
B) the bottleneck effect
C) the founder effect
D) gene flow
E) stabilising selection

Section D - Quantitative Chemistry (Q31-44)

- 31.** Sulfur forms sulfuric acid, H_2SO_4 . Which of the following correctly gives both the oxidation number of sulfur in H_2SO_4 and the position of sulfur in the periodic table?
- A)** -2 ; Period 4, Group 16
B) $+2$; Period 2, Group 16
C) $+4$; Period 3, Group 16
D) $+6$; Period 3, Group 17
E) $+6$; Period 3, Group 16
- 32.** What is the oxidation number of manganese in KMnO_4 ?
- A)** $+2$ **B)** $+4$ **C)** $+6$ **D)** $+7$ **E)** $+8$
- 33.** Zinc metal is placed in copper(II) sulfate solution: $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$. Which statement about this reaction is correct?
- A)** Cu metal acts as the reducing agent
B) Zn^{2+} is reduced because it loses electrons
C) Cu^{2+} is reduced because it gains electrons
D) Zn is reduced because it gains electrons
E) Cu^{2+} is oxidised because it loses electrons
- 34.** What is the hydrogen ion concentration in a solution of pH 4.0?
- A)** $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
B) $1.0 \times 10^{-3} \text{ mol dm}^{-3}$
C) $1.0 \times 10^4 \text{ mol dm}^{-3}$
D) $1.0 \times 10^{-4} \text{ mol dm}^{-3}$
E) $4.0 \times 10^{-1} \text{ mol dm}^{-3}$
- 35.** At 25°C , $K_w = 1.0 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. A solution has $[\text{OH}^-] = 1.0 \times 10^{-4} \text{ mol dm}^{-3}$. What is the pH of this solution?
- A)** 4 **B)** 7 **C)** 10 **D)** 11 **E)** 14
- 36.** What is the pH of a $0.001 \text{ mol dm}^{-3}$ solution of hydrochloric acid, a strong monoprotic acid?
- A)** -3 **B)** 0.001 **C)** 2 **D)** 3 **E)** 11
- 37.** How many molecules are present in 22.4 dm^3 of oxygen gas at STP? (Molar volume at STP = $22.4 \text{ dm}^3 \text{ mol}^{-1}$; Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)
- A)** 1.204×10^{24} **B)** 1.35×10^{25} **C)** 3.01×10^{23} **D)** 6.02×10^{23} **E)** 6.02×10^{22}
- 38.** 0.5 mol of sodium hydroxide is dissolved in water and the solution is made up to 250 cm^3 . What is the concentration of the solution in mol dm^{-3} ?
- A)** $0.125 \text{ mol dm}^{-3}$ **B)** 0.5 mol dm^{-3} **C)** 1.25 mol dm^{-3} **D)** 2.0 mol dm^{-3} **E)** 5.0 mol dm^{-3}
- 39.** Which law states that the volume of a gas is directly proportional to the number of moles of gas present, provided the temperature and pressure are constant?
- A)** Avogadro's law
B) Boyle's law
C) Charles's law
D) Gay-Lussac's law
E) Dalton's law of partial pressures
- 40.** A gas occupies 9.0 dm^3 at 27°C and constant pressure. It is then cooled to -73°C , still at the same pressure. What volume does it now occupy?
- A)** -24.3 dm^3 **B)** 3.0 dm^3 **C)** 6.0 dm^3 **D)** 13.5 dm^3 **E)** 24.3 dm^3
- 41.** A gas occupies 500 cm^3 at a pressure of 100 kPa , the temperature remaining constant. The pressure is increased to 250 kPa . What is the new volume?
- A)** 50 cm^3 **B)** 100 cm^3 **C)** 200 cm^3 **D)** 250 cm^3 **E)** 1250 cm^3

42. Each of the following compounds contains two carbon atoms. Which has the highest boiling point?
A) CH_3CH_3 (ethane)
B) CH_3COOH (ethanoic acid)
C) CH_3OCH_3 (methoxymethane)
D) CH_3CHO (ethanal)
E) $\text{CH}_3\text{CH}_2\text{OH}$ (ethanol)
43. Which one of the following molecules is polar?
A) NH_3 B) BF_3 C) CO_2 D) CCl_4 E) BeCl_2
44. Given the following data: $\text{C(s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$, $\Delta H = -394 \text{ kJ mol}^{-1}$; $\text{CO(g)} + \frac{1}{2}\text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$, $\Delta H = -283 \text{ kJ mol}^{-1}$. What is ΔH for the reaction $\text{C(s)} + \frac{1}{2}\text{O}_2\text{(g)} \rightarrow \text{CO(g)}$?
A) -677 kJ mol^{-1} B) -394 kJ mol^{-1} C) -283 kJ mol^{-1} D) -111 kJ mol^{-1} E) $+111 \text{ kJ mol}^{-1}$

Section E - Mathematics (Q45-53)

45. Solve for x : $\sqrt{3x - 2} = 4$.
A) $x = 18$ B) $x = 14/3$ C) $x = 4$ D) $x = 2$ E) $x = 6$
46. Solve for x : $\sqrt{x + 6} = x$. Which answer lists ALL valid solutions?
A) $x = 6$
B) $x = 3$ only
C) $x = -2$ only
D) $x = 3$ and $x = -2$
E) $x = 2$
47. Solve the inequality: $5 - 3x \geq 20$.
A) $x \geq -5$ B) $x \leq -5$ C) $x \geq 5$ D) $x < -5$ E) $x \leq 5$
48. Solve the inequality: $-2x + 7 > 15$.
A) $x < -4$ B) $x < 4$ C) $x > -4$ D) $x > 4$ E) $x < -11$
49. If $f(x) = 3x^2 - 4x + 5$, what is $f(2)$?
A) 4 B) 9 C) 13 D) 25 E) 33
50. A 15 m ladder leans against a vertical wall with its foot 9 m from the base of the wall. How high up the wall does the ladder reach?
A) 6 m B) 12 m C) 13 m D) 17.5 m E) 24 m
51. A rectangle has a width of 16 cm and a diagonal of 34 cm. What is its perimeter?
A) 46 cm B) 62 cm C) 92 cm D) 100 cm E) 128 cm
52. Solve for x : $4(x - 2) = 2x + 2$.
A) $x = 4$ B) $x = -3$ C) $x = 2$ D) $x = 10$ E) $x = 5$
53. Using $\pi = 3.14$, what is the area of a circle with a diameter of 12 cm?
A) 37.68 cm^2 B) 75.36 cm^2 C) 113.04 cm^2 D) 226.08 cm^2 E) 452.16 cm^2

Section F - Physics (Q54-62)

54. A car travels at 72 km/h. What is this speed in m/s?
A) 7.2 m/s B) 20 m/s C) 24 m/s D) 36 m/s E) 240 m/s
55. A cyclist rides at 15 m/s. What is this speed in km/h?
A) 4.2 km/h B) 5 km/h C) 45 km/h D) 54 km/h E) 150 km/h
56. Two 6Ω resistors are connected in parallel. What is the total resistance?
A) $1/3 \Omega$ B) 3Ω C) 6Ω D) 12Ω E) 36Ω

57. Three resistors of $4\ \Omega$, $4\ \Omega$ and $2\ \Omega$ are connected in parallel. What is the total resistance?
A) $0.25\ \Omega$ **B)** $1\ \Omega$ **C)** $2\ \Omega$ **D)** $3.3\ \Omega$ **E)** $10\ \Omega$
58. A current of $3\ \text{A}$ flows through a $4\ \Omega$ resistor. What power is dissipated?
A) $7\ \text{W}$ **B)** $12\ \text{W}$ **C)** $36\ \text{W}$ **D)** $48\ \text{W}$ **E)** $144\ \text{W}$
59. An appliance draws a current of $2.5\ \text{A}$ when connected to a $12\ \text{V}$ supply. What power does it use?
A) $0.21\ \text{W}$ **B)** $4.8\ \text{W}$ **C)** $14.5\ \text{W}$ **D)** $30\ \text{W}$ **E)** $60\ \text{W}$
60. What is the gauge pressure at a depth of $5\ \text{m}$ in water? ($\rho = 1000\ \text{kg/m}^3$, $g = 9.8\ \text{m/s}^2$)
A) $4,900\ \text{Pa}$ **B)** $9,800\ \text{Pa}$ **C)** $49,000\ \text{Pa}$ **D)** $49\ \text{Pa}$ **E)** $490,000\ \text{Pa}$
61. A block has a mass of $540\ \text{g}$ and a volume of $200\ \text{cm}^3$. What is its density, and which metal is it consistent with?
A) $0.37\ \text{g/cm}^3$ — a light alloy
B) $2.7\ \text{g/cm}^3$ — aluminium
C) $27\ \text{g/cm}^3$ — a heavy alloy
D) $108\ \text{g/cm}^3$ — gold
E) $740\ \text{g/cm}^3$ — lead
62. A lamp produces a light intensity of $90\ \text{W/m}^2$ at a distance of $1\ \text{m}$. What is the intensity at $3\ \text{m}$ from the lamp?
A) $10\ \text{W/m}^2$ **B)** $30\ \text{W/m}^2$ **C)** $90\ \text{W/m}^2$ **D)** $270\ \text{W/m}^2$ **E)** $810\ \text{W/m}^2$

Section G - Logic & Reasoning (Q63-70)

63. 'If Maria catches the 8:00 train, she arrives before noon.' Maria did not arrive before noon. Which conclusion necessarily follows?
A) If Maria does not catch the 8:00 train, she does not arrive before noon.
B) If Maria arrives before noon, she caught the 8:00 train.
C) Maria missed the 8:00 train because she overslept.
D) Maria caught the 8:00 train, but the train was delayed.
E) Maria did not catch the 8:00 train.
64. A technician states: 'If the sample is contaminated, the result is unreliable.' Which statement, if true, contradicts this?
A) The sample is contaminated and the result is reliable.
B) The sample is not contaminated and the result is unreliable.
C) The sample is contaminated and the result is unreliable.
D) If the sample is not contaminated, the result is reliable.
E) The sample is not contaminated and the result is reliable.
65. Anna reasons: 'If the alarm rings, the neighbours call the police. The alarm did not ring, so the neighbours did not call the police.' Her reasoning is...
A) invalid: the original conditional itself must be false because the neighbours might call anyway
B) valid: it correctly applies the contrapositive of the original statement
C) valid: the inverse of a conditional always follows from it
D) valid: the neighbours can only call the police when the alarm rings
E) invalid: it denies the antecedent, and the inverse is not equivalent to the original conditional
66. All cardiologists are doctors. Some teachers are doctors. No cardiologist is a teacher. Which conclusion follows necessarily?
A) All teachers are doctors.
B) All doctors are either cardiologists or teachers.
C) No teacher is a doctor.
D) Some cardiologists are teachers.
E) Some teachers are not cardiologists.

- 67.** No reptiles are warm-blooded. All snakes are reptiles. Which conclusion follows necessarily?
- A)** No snakes are warm-blooded.
 - B)** All warm-blooded animals are reptiles.
 - C)** Some snakes are warm-blooded.
 - D)** No animal other than reptiles is warm-blooded.
 - E)** Some reptiles are warm-blooded.
- 68.** A study of 5,000 cyclists found that riders without helmets were more than three times as likely to sustain head injuries as riders wearing helmets. Which conclusion is best supported?
- A)** In this study, unhelmeted riders had a higher incidence of head injury than helmeted riders.
 - B)** Cycling without a helmet causes riders to crash more often.
 - C)** Unhelmeted riders take more risks than helmeted riders.
 - D)** Helmets prevent all head injuries in cycling accidents.
 - E)** Helmet laws should be introduced to reduce all cycling injuries.
- 69.** A survey of 2,000 employees found that those who walk to work reported, on average, higher job satisfaction than those who commute by car. Which conclusion follows?
- A)** Car commuters would become more satisfied if they switched to walking.
 - B)** Employers should subsidise walking programmes in order to raise productivity.
 - C)** Walking to work causes greater job satisfaction.
 - D)** Every employee who walks to work is more satisfied than every employee who drives.
 - E)** Among surveyed employees, walking commuters reported higher average satisfaction than car commuters.
- 70.** What is the angle between the hour hand and the minute hand of an analogue clock at 3:15?
- A)** 0 degrees **B)** 7.5 degrees **C)** 15 degrees **D)** 30 degrees **E)** 82.5 degrees

Answer key

This key refers to the option order printed in this paper. The online app shuffles option order on every attempt, so answer letters inside the app will differ from the ones here.

1	C	2	E	3	D	4	A	5	B
6	A	7	E	8	D	9	C	10	B
11	C	12	E	13	A	14	D	15	B
16	B	17	D	18	E	19	C	20	A
21	C	22	D	23	B	24	A	25	E
26	C	27	D	28	A	29	E	30	B
31	E	32	D	33	C	34	D	35	C
36	D	37	D	38	D	39	A	40	C
41	C	42	B	43	A	44	D	45	E
46	B	47	B	48	A	49	B	50	B
51	C	52	E	53	C	54	B	55	D
56	B	57	B	58	C	59	D	60	C
61	B	62	A	63	E	64	A	65	E
66	E	67	A	68	A	69	E	70	B